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Safety-Rated EPS Torque Sensor Interface PCB

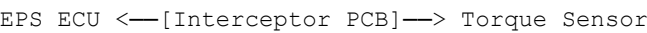
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1. OBJECTIVE

This document records the complete hardware design study for a PCB that intercepts the torque sensor signal between the Perodua EPS ECU and its torque sensor, enabling KommuAssist2 to provide Level 2 ADAS steering assistance.

The PCB sits physically in the wire harness between the ECU and the torque sensor:



The default and safest state is OEM pass-through. Any fault, uncertainty, timeout, power issue, software error, signal mismatch, or diagnostic failure shall return the EPS ECU to the original torque sensor signal without delay.

NOTE: This document covers a study and prototype phase only. Public-road deployment requires functional safety assessment, legal review, vehicle-level validation, and regulatory approval per the requirements document.

2. SYSTEM OVERVIEW

2.1 Core Concept

The interceptor PCB operates in two modes :

- 1. OEM pass-through (default) : The torque sensor signal passes directly through the PCB to the ECU unchanged. This is the hardware default with no firmware required.
- 2. Boost active : The MCU modifies the torque signal within safe bounds when requested by KommuAssist2 via CAN bus, subject to supervisor approval.

2.2 Confirmed Signal Characteristics

Parameter	Value	Source
Sensor type		

2.3 Power Architecture

The PCB is powered from the ECU's 5V VDD pin and not from the 12V battery due to simpler design, and no need for 12V step-down converter

- Cleaner power, already regulated by ECU
- PCB taps a small amount of ECU 5V while passing the full 5V through to the sensor unchanged
- 5V is regulated down to 3.3V on-board via AMS1117-3.3 LDO for all PCB chips

3. SAFETY ARCHITECTURE

3.1 Safety-Partitioned Interposer

The design implements Concept C from the requirements document, the most robust of the three recommended architectures :

- Control path: Main MCU (STM32G0B1CBT6) calculates boost values and drives DAC outputs
- Safety path: Independent supervisor MCU (STM32G030F6P6TR) independently monitors all signals and can force pass-through without relying on main MCU firmware
- Hardware enforcement: AND gate (SN74LVC1G08) ensures both supervisor signals must be HIGH simultaneously for boost to activate and cannot be bypassed by software
- Hardware default: TS5A23157 NC analog switch defaults to OEM pass-through with zero power with no firmware needed

3.1 Safety Goals Addressed

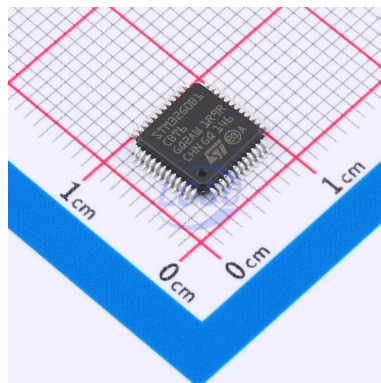
Safety Goal	How addressed in design
Prevent unintended steering assist	Dual gate logic : AND gate requires both FORCE_PT and GATE_ENABLE HIGH
Prevent loss of driver steering authority	NC switch default : OEM always available hardware path
Prevent sustained wrong-direction assist	LM393 output monitor detects mismatch → will immediately force pass-through

Prevent modification when vehicle state unknown	CAN timeout detection in supervisor firmware
Prevent modification during fault	Supervisor forces FORCE_PT=LOW on any fault condition
Immediate reversion on fault	Hardware AND gate : pull either input LOW → instant pass-through
No unsafe behavior during startup/reset	BOOT0 pull-down ensures clean boot, NC switch active during startup
No single fault causing sustained wrong output	Dual gate + independent supervisor + LM393 hardware monitor
Diagnostic fault logging	

3.3 Non-negotiable Safety Design Compliance

4. COMPONENT SELECTION RATIONALE

4.1 Main MCU : STM32G0B1CBT6 (LQFP-48)

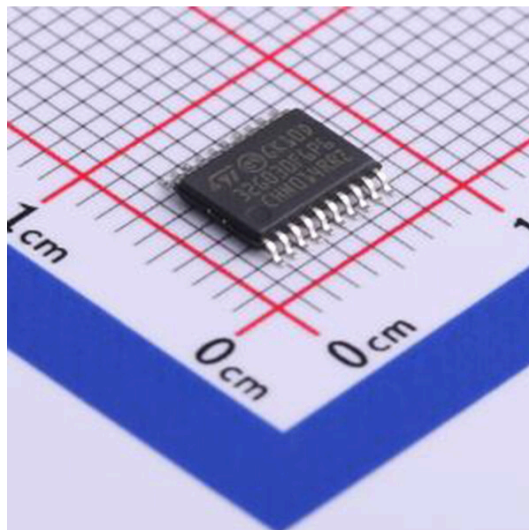


Parameter	Detail
Part Number	STM32G0B1CBT6
Package	LQFP - 48
Core	
Flash	
RAM	
Supply Voltage	
CAN Standard	
Price	\$2.22 per unit

Why STM32G0B1CBT6:

- Dual DAC outputs (PA4 and PA5) : Needed for independent MAIN and SUB signal generation
- FDCAN peripheral : Required for vehicle CAN bus communication
- Sufficient ADC channels for reading MAIN_ADC and SUB_ADC
- 3.3V operation : Matches system power rail
- LQFP-48 package : Solderable by hand for prototype
- Good availability on LCSC and JLCPCB

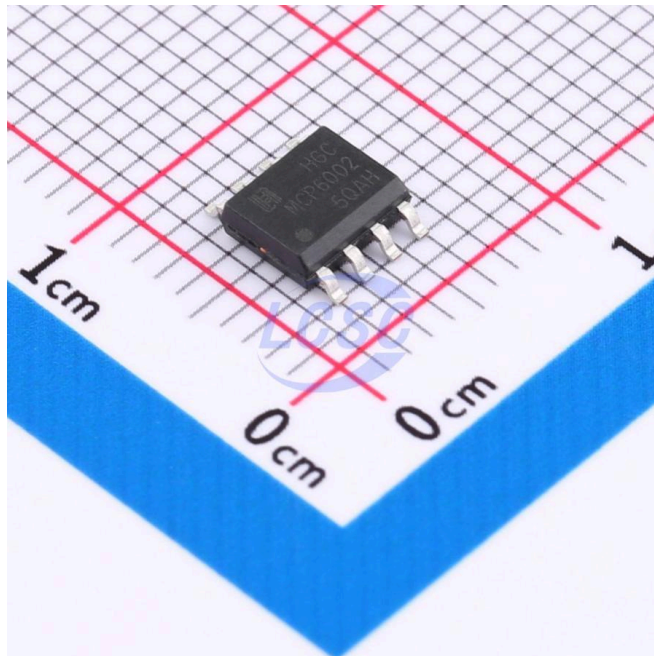
4.2 Safety Supervisor : STM32G030F6P6TR (TSSP-20)



Parameter	Detail
Part Number	STM32G030F6P6TR

Package	TSSP-20
Core	
Flash	
RAM	
Supply Voltage	
CAN Standard	
Price	\$2.22 per unit

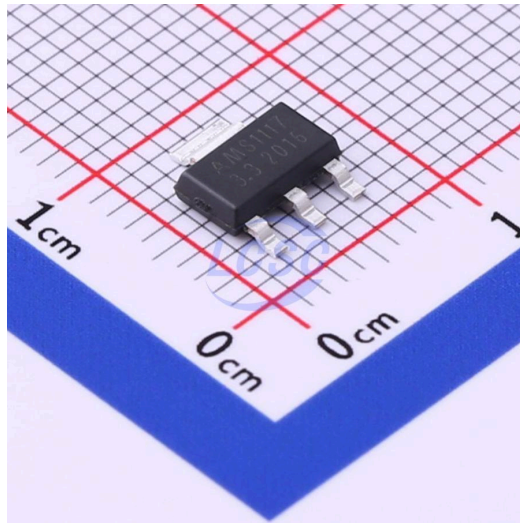
4.3 Op-amp : MCP6002DRG (SOP-8, Dual)



Why MCP6002:

- Rail-to-rail output : Can output very close to 0V and very close to VCC (5V), essential for full 0–5V signal range
- Single supply operation : Works directly from VCC without negative rail
- Dual channel in one package : One chip handles both MAIN and SUB channels, reduces component count
- Low offset voltage : Accurate enough for analog torque signal conditioning
- Used on two sheets : Same chip in BOM simplifies ordering

4.4 LDO Regulator : AMS1117



Parameter	Detail
Part Number	AMS1117
Package	
Input Voltage	
Output Voltage	
Max Current	
Price	

Why AMS1117 (LDO) not a buck converter:

- Small voltage drop ($5V \rightarrow 3.3V$, only 1.7V gap) where LDO is efficient enough
- Low current draw ($\sim 100\text{-}200\text{mA}$ total) : No need for high-efficiency switching
- No switching noise : buck converters generate high-frequency noise that would corrupt sensitive analog torque signals
- Simpler circuit : Fewer components, easier to prototype
- Built-in thermal shutdown protection

Buck Converter has a lot of rapid switches, can lead to current drop/loss and kacau signal hence tak selamat and tak accurate.